MH1100 Calculus I

Handout 2 - Limits I

Synopsis

Introduction to limit

- What is a limit? A key example
- Limit and the tangent to a curve, limit and the velocity of an object
- Intuitive definition of limit
- One-sided limits, infinite limits, vertical asymptote

Limits

Goal

Our main task in this lecture is to start thinking about limits.

This is one of the most fundamental concepts in calculus (and, in fact, is one of the most important ideas in all of mathematics).

Notation

Here is how we denote a limit:

$$\lim_{x \to a} f(x)$$

You read this expression in the following way: The limit of the function $f(x)$ as x approaches a.

Limits

Limit is one of the most fundamental concepts in calculus.

Precise Definition of a Limit

Let $f(x)$ be a function defined on some open interval that contains a, except possibly at a itself. Then we say that **the limit of $f(x)$ as x approaches a is L**, and we write

$$\lim_{x \to a} f(x) = L$$

if for every number $\epsilon > 0$ there is a number $\delta > 0$ such that

$$\text{if} \quad 0 < |x - a| < \delta \quad \text{then} \quad |f(x) - L| < \epsilon$$

Definition: Continuity

Let f be a function and let $a \in \mathbb{R}$. The function f is said to be continuous at the point a if the following three things are true:

- $f(a)$ is defined (that is, a is in the domain of f).
- The limit $\lim_{x \to a} f(x)$ exists.
- $\lim_{x \to a} f(x) = f(a)$.

Limits

Limit is one of the most fundamental concepts in calculus.

Definition: The derivative at a

The derivative of a function f at a number a, denoted by $f'(a)$, is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

if this limit exists.

Definition: Definite Integral

If f is a function defined for $a \le x \le b$, we divide the interval $[a, b]$ into n subintervals of equal width $\Delta x = (b-a)/n$. We let $x_0(=a), x_1, x_2, \ldots, x_n(=b)$ be the endpoints of these subintervals and we let $\xi_1, \xi_2, \ldots, \xi_n$ be any **sample points** in these subintervals, so ξ_i lies in the ith subinterval $[x_{i-1}, x_i]$. Then the **definite integral of f from a to b** is

$$\int_a^b f(x)dx = \lim_{n \to \infty} \sum_{i=1}^{n} f(\xi_i)\Delta x$$

provided that this limit exists and gives the same value for all possible choices of sample points. If it does exist, we say that f is **integrable** on $[a, b]$.

What is a limit? A key example

Let's think about the function

$$f(x) = \frac{\sin x}{x},$$

not defined here

and focus on how it looks close to the point $\underline{x = 0}$.

Question. What is the domain of $f(x)$?

Answer. $\mathbb{R} \setminus \{0\}$.

Question. What is the value of $f(0)$? not defined.

Answer. The point 0 does not lie in the domain of f, so f does not assign any value to the point $x = 0$.

What is a limit? A key example (Continued)

Let's think about the function

$$f(x) = \frac{\sin x}{x},$$

and focus on how it looks close to the point $x = 0$.

draw graphs

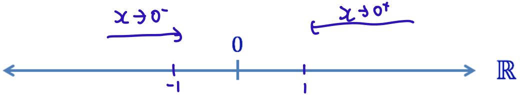
The key idea

Even though $f(x) = \frac{\sin x}{x}$ has no value defined at the point $x = 0$, we can still look at the values f takes at points close to $x = 0$, and see if those values are approaching some fixed value as x gets closer and closer to 0.

That is precisely what the limit of a function $f(x)$ at a point a is: a number L which the values of the function get "closer and closer" to as the variable x gets "closer and closer" to a.

What is a limit? A key example (Continued)

Let's calculate some values and see what happens as the variable x gets close to $x = 0$:



Approaching 0 from the left: $x \to 0^-$

x	-0.5	-0.25	-0.1	-0.05	-0.01	-0.005	-0.001
$\frac{\sin x}{x}$							

→ gets closer and closer to zero

Approaching 0 from the right:

x	0.5	0.25	0.1	0.05	0.01	0.005	0.001
$\frac{\sin x}{x}$							

What is a limit? A key example (Continued)

Approaching 0 from the left:

$f(x) = \frac{\sin x}{x}$ is an even function. $f(-x) = f(x)$
→ so both tables have same values

x	-0.5	-0.25	-0.1	-0.05	-0.01	-0.005	-0.001
$\frac{\sin x}{x}$	0.9589	0.9896	0.9983	0.9996	0.999983	0.999996	0.999998

getting closer to 1

Approaching 0 from the right:

x	0.5	0.25	0.1	0.05	0.01	0.005	0.001
$\frac{\sin x}{x}$	0.9589	0.9896	0.9983	0.9996	0.999983	0.999996	0.999998

getting closer to 1

Conclusion: $\lim_{x\to 0} \frac{\sin x}{x} = ?$ 1

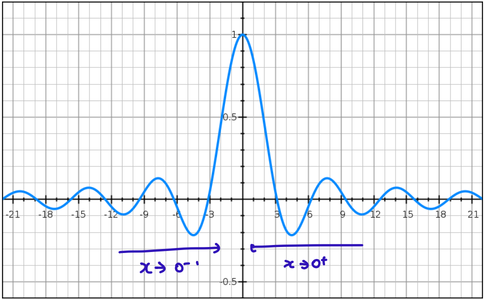
Comment: At this point, this is only a guess. We cannot say for sure what this limit is until we understand exactly what a limit is.

We will meet a logically rigorous definition of *limit* later on, and after that we will be able to prove equations like this with certainty.

Limits

The graph of function

$$f(x) = \frac{\sin x}{x}$$



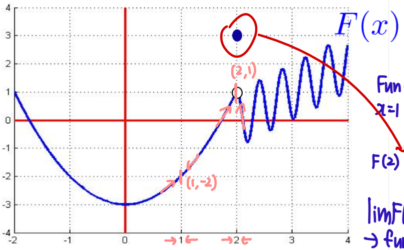
As shown in the above figure, we can also guess that

$$\lim_{x \to 0} f(x) = \lim_{x \to 0} \frac{\sin x}{x} = 1.$$

Example A

The graph of the function $F(x)$ is shown below. Estimate the limits $\lim_{x \to 1} F(x)$ and $\lim_{x \to 2} F(x)$.

-2 $\qquad$ 1



Function is defined @ $x=1$ and $x=2$

$F(2)=3$

$\lim F(x) \neq F(2)$

→ function is not continuous at $x=2$.

Comment: This shows that a limit of a function $\lim_{x \to x_0} F(x)$ at some point x_0 has nothing to do with $F(x_0)$, the actual value of the function at that point. Instead, a limit represents how the function looks "close" to x_0.

Example B

Estimate $\lim_{x\to 0} F(x)$, where $F(x)$ is defined by the rule

$$F(x) = \begin{cases} 1 + \sin\frac{1}{x}, & x > 0 \\ 1, & x \leq 0 \end{cases}$$

$2n\pi < \frac{1}{x} \leq 2(n+1)\pi \quad (n \in \mathbb{Z}, n > 0)$

$-1 \leq \sin\frac{1}{x} \leq 1$

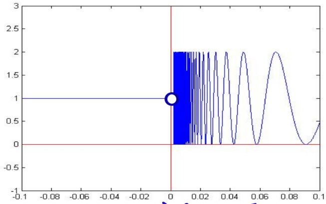
$0 \leq 1 + \sin\frac{1}{x} \leq 2$

$\frac{1}{2n\pi} > x \geq \frac{1}{2(n+1)\pi}$

$\rightarrow$ if $x \to 0$, $n \to \infty$

If $x > 0$, then $\frac{1}{x} > 0$.

$\lim_{x\to 0} F(x)$ does not exist



$x \to 0^-$ $\quad$ $x \to 0^+$

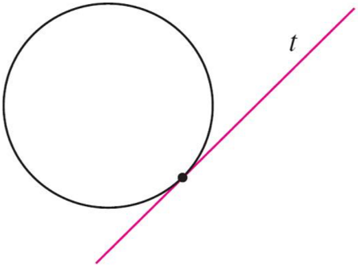
$F(0) = 1$

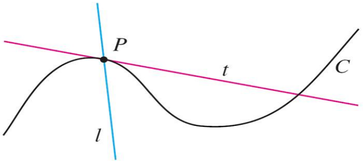
Comment: This is because as x approaches 0 from the right, the value $f(x)$ does not approach any single value.

Limit and the tangent to a curve

A tangent to a curve

A tangent to a curve is a line that touches the curve. In other words, a tangent line should have the same direction as the curve at the point of contact.

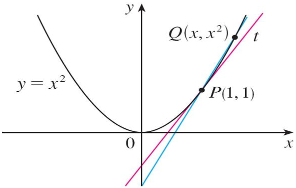




Example 1

Find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.

(x_0, y_0) gradient : m
$y - y_0 = m(x - x_0)$



$m_{PQ} = \frac{x^2 - 1}{x - 1} \rightarrow$ not gradient of tangent!

* If Q is closer to P, it would be a better approximation.

Solution:

gradient

We will be able to find an equation of the tangent line t as soon as we know its slope m.

We can compute an approximation to m by choosing a nearby point $Q(x, x^2)$ on the parabola and computing the slope m_{PQ} of the secant line PQ. *A secant line is a line that cuts (intersects) a curve more than once.*

Example 1–Solution

We choose $x \neq 1$ so that $P \neq Q$. Then $m_{PQ} = \frac{x^2-1}{x-1}$. $= \frac{(x+1)(x-1)}{x-1}$

$= x+1$

For instance, for the point $Q(1.5, 2.25)$ we have

$$m_{PQ} = \frac{x^2-1}{x-1} = \frac{2.25-1}{1.5-1} = \frac{1.25}{0.5} = 2.5.$$

The tables below show the values of m_{PQ} for several values of x close to 1.

- Approaching 1 from the right:

x	2	1.5	1.1	1.01	1.001
m_{PQ}	3	2.5	2.1	2.01	2.001

approaching 2

- Approaching 1 from the left:

x	0	0.5	0.9	0.99	0.999
m_{PQ}	1	1.5	1.9	1.99	1.999

approaching 2.

The closer Q is to P, the closer x is to 1 and, it appears from the tables, the closer m_{PQ} is to 2.

Example 1–Solution

The closer Q is to P, the closer m_{PQ} is to 2. This suggests that the slope of the tangent line t should be $m = 2$.

We say that the slope of the tangent line is the limit of the slopes of the secant lines, and we express this symbolically by writing

$$\lim_{Q \to P} m_{PQ} = m \qquad \text{and} \qquad \lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2.$$

Slope of the tangent line is the limit.

$=2$

Assuming that the slope of the tangent line is indeed 2, we use the point-slope form of the equation of a line $[y - y_1 = m(x - x_1)]$ to write the equation of the tangent line through $(1, 1)$ as

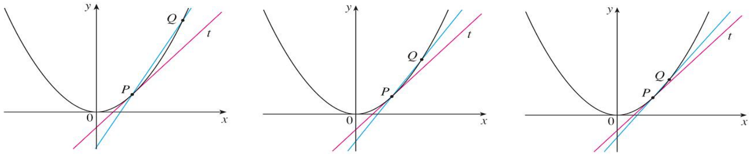
$$y - 1 = 2(x - 1) \qquad \text{or} \qquad y = 2x - 1.$$

Example 1–Solution

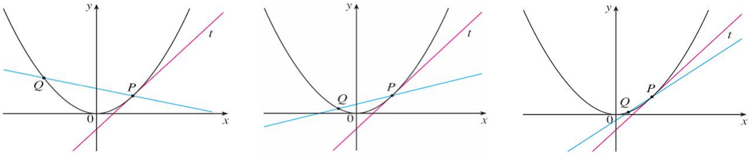
Figures below illustrate the limiting process that occurs in this example.

When Q move closer to P, the secant line becomes a better approximation for the tangent line.

Q approaches *P* from the right



Q approaches *P* from the left



this part i not sure if i heard correctly

D is a subset of real numbers, and is a domain of function f.

Definition

Let $D \subset \mathbb{R}$ and let $f : D \to \mathbb{R}$ be a function. Then we define the tangent line to the graph of $f(x)$ at the point x_0 to be the line going through the point $(x_0, f(x_0))$ with gradient

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0) \to \text{if the limit exists.}$$

if the limit exists. If this limit does not exist then we say that the graph does not have a tangent at that point.

Exercise

$0^0 \to$ not defined

Find an equation of the tangent line to the function $y = x^x$ at the point $P(1,1)$. $Q(x, x^x)$

$y - 1 = m(x-1)$ $\quad m_{PQ} = \dfrac{x^x - 1}{x-1}$

$\lim_{x \to 1} m_{PQ} = m$

x	2	1.5	1.1	1.01
m_{PQ}	3	1.67	1.10	1.010

$\longrightarrow 1$

x	0.1	0.5	0.9	0.99	0.999
m_{PQ}	0.228	0.585	0.904	0.99	0.999

$\longrightarrow 1$

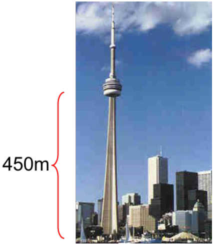
both approaches 1

$\therefore$ Eqn of tangent line: $y = x$

Limit and the velocity of an object

Around 1590, Galilleo made the empirical discovery that if an object is dropped and falls freely, then the distance $s(t)$ it has fallen after t seconds is approximately given by the law:

$$s(t) = 4.9t^2 = \frac{1}{2}gt^2 \qquad v(t) = gt$$



Should still be in the air.

Example 3

Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground. Find the velocity of the ball after 5 seconds.

Example 3–Solution

Solution:

The difficulty in finding the instantaneous velocity after 5 seconds is that we are dealing with a single instant of time ($t = 5$), so no time interval is involved.

We can approximate the desired quantity by computing the average velocity over the brief time interval of τ seconds starting at time $t = 5$

can take a negative value

$$\begin{aligned} \text{average velocity} &= \frac{\text{change in position}}{\text{time elapsed}} \\ &= \frac{s(5+\tau) - s(5)}{\tau} \\ &= \frac{4.9(5+\tau)^2 - 4.9 \times 5^2}{\tau} \\ &= 49 + 4.9\tau \end{aligned}$$

$t = 5 + \tau$ ($\tau < 0$)

$t = 5$

$$V_{ave} = \frac{s(5) - s(5+\tau)}{-\tau} = \frac{s(5+\tau) - s(5)}{\tau}$$

$t=5$, $t=5+\tau$, $s(t)$, $s(t+\tau)$

As the time interval τ gets smaller and smaller, this quantity should become a better and better approximation of the instantaneous velocity at time t.

Thus, the instantaneous velocity when $t = 5$ can be defined to be the limiting value of these average velocities over shorter and shorter time periods that start at $t = 5$. It appears that the (instantaneous) velocity after 5 seconds is $v = 49$ m/s.

Limit and the velocity of an object

Definition

Assume that a particle's trajectory (in one dimension) is described by a function $s : \mathbb{R} \to \mathbb{R}$. The instantaneous velocity of the particle at time t is defined to be:

$$\lim_{\tau \to 0} \frac{s(t+\tau) - s(t)}{\tau}.$$

Examples 1 and 3 show that in order to solve tangent and velocity problems we must be able to find limits.

But what is a limit?

$\lim_{x \to a} f(x)$

The limit of a function

The precise definition of 'limit' is the most challenging idea in this course, so we will come back to it once we have developed some feeling for what a limit actually is. For the time being we will use the following.

Intuitive Definition of a Limit

Suppose $f(x)$ is defined when x is near the number a. (This means that f is defined on some open interval that contains a, except possibly at a itself.) Then we write

$(a-\delta, a+\delta)\setminus\{a\}$
$\delta>0$

$$\lim_{x\to a} f(x) = L$$

and say *"the limit of $f(x)$, as x approaches a, equals L"*

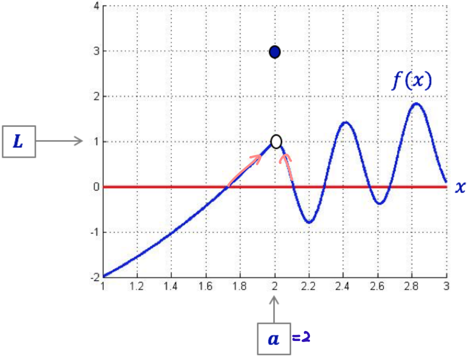
if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by restricting x to be sufficiently close to a (on either side of a) but not equal to a.

Comment: This is actually the full correct definition of limit, but stated using language that is insufficiently precise for mathematics.

An example illustrating our intuitive definition of limit

We expect $\lim_{x \to 2} f(x) = ?$

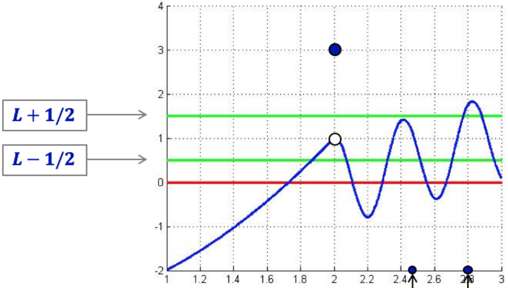
$= 1$



Question: How “close” to $a = 2$ must we restrict the variable x to ensure that the corresponding values $f(x)$ lie within $\pm\frac{1}{2}$ of the proposed limit $L = 1$?

An example illustrating our intuitive definition of limit (cont.)

To begin: note that a value $f(x)$ lies "within $\pm\frac{1}{2}$ of the proposed limit $L = 1$" precisely when the corresponding point on the graph lies within the two green lines in the graph below:



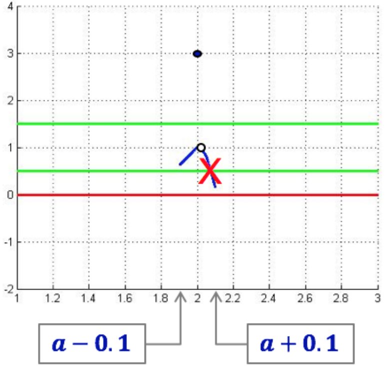
Open interval cannot be from 1.8 to 2.2.
→ values are outside the green line
→ $(1.8, 2.2) \setminus \{2\}$ X → doesn't work

E.g. At this x, $f(x)$ lies within $\pm\frac{1}{2}$ of $L = 1$...

...but at this x, $f(x)$ *does not* lie within $\pm\frac{1}{2}$ of $L = 1$.

An example illustrating our intuitive definition of limit (cont.)

Question: If we restrict x to within ± 0.1 of $a = 2$, is $f(x)$ within $\pm\frac{1}{2}$ of $L = 1$? (Except at the point a itself.)



Answer: Not necessarily. There is still a part outside the two green lines

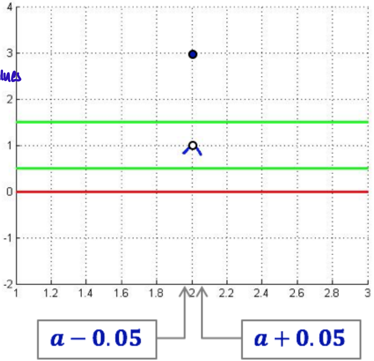
An example illustrating our intuitive definition of limit (cont.)

Question: If we restrict x to within ± 0.05 of $a = 2$, is $f(x)$ within $\pm\frac{1}{2}$ of $L = 1$? (Except at the point a itself.)

$\lim_{x \to 2} f(x) = 1$

$\varepsilon > 0$ dist between green line and L

need to find the open interval for various ε values to prove $\lim_{x \to 2} f(x) = 1$



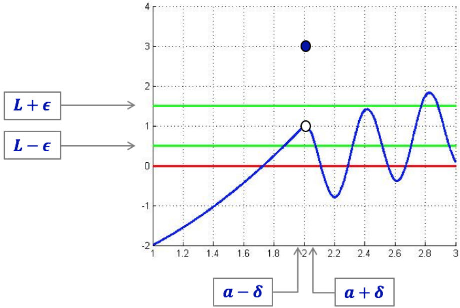
$(1.95, 2.05) \setminus \{2\} \to$ will not work if we limit x to within ± 0.025 of $a=2$.

Answer: Yes.

An example illustrating our intuitive definition of limit (cont.)

If $-\frac{1}{2} < f(x) - L < \frac{1}{2}$, $L=1$, then $f(x)$ is between the 2 green lines $y = 1 - \frac{1}{2}$ and $y = 1 + \frac{1}{2}$

Conclusion of this example

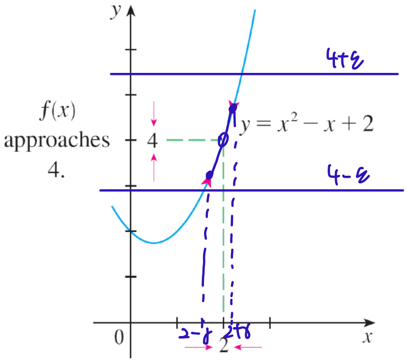


In this example, $\lim_{x \to 2} f(x) = 1$, because it doesn't matter how small we make this distance ϵ, we can always find a distance δ such that the part of the graph which lies over $(a-\delta, a+\delta) \setminus \{a\}$ lies entirely between the green lines.

$(L-\varepsilon, L+\varepsilon)$ $\varepsilon > 0$

The definition of a limit says that the values of $f(x)$ approach L as x approaches a. In other words, the values of $f(x)$ tend to get closer and closer to the number L as x gets closer and closer to the number a (from either side of a) but $x \neq a$.

$\lim_{x \to 2} (x^2 - x + 2) = 4$



$\varepsilon > 0$

$L+\varepsilon$
$L-\varepsilon$
$L=4$

The limit of a function

An alternative notation for

$$\lim_{x \to a} f(x) = L$$

is

$$f(x) \to L \quad \text{as} \quad x \to a$$

which is usually read $f(x)$ *approaches* L *as* x *approaches* a.

Notice the phrase "but $x \neq a$" in the definition of limit. This means that in finding the limit of $f(x)$ as x approaches a, we never consider $x = a$. In fact, $f(x)$ needs not even be defined when $x = a$. The only thing that matters is how f is defined near a.

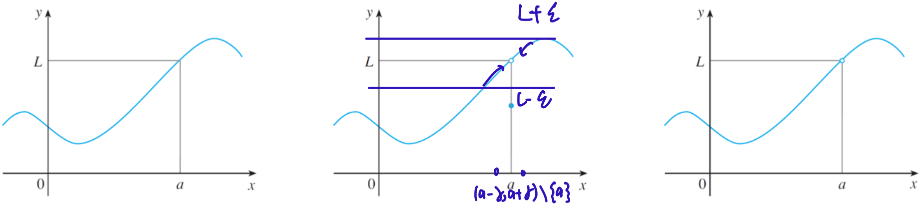
$\hookrightarrow x \in (a-\delta, a+\delta) \setminus \{a\}$

The figure below shows the graphs of three functions. Note that in part (c), $f(a)$ is not defined and in part (b), $f(a) \neq L$.

But in each case, regardless of what happens at a, it is true that $\lim_{x \to a} f(x) = L$.

$f(a) \neq L$

$f(a)$ not defined.



(a) (b) (c)

$\lim_{x \to a} f(x) = L$ $\lim_{x \to a} f(x) = L$ $\lim_{x \to a} f(x) = L$

Example 1

Guess the value of $\lim_{x\to 1} \frac{x-1}{x^2-1}$ Domain of $f = \mathbb{R} \setminus \{x=1, x=-1\}$

Solution

The function $f(x) = (x-1)/(x^2-1)$ is not defined when $x = 1$, but that doesn't matter because we consider values of x that are close to a but not equal to a.

The tables below give values of $f(x)$ (correct to six decimal places) for values of x that approach 1 (but are not equal to 1).

Approaching 1 from the left:

$x < 1$	0.5	0.9	0.99	0.999	0.9999
$f(x)$	0.666667	0.526316	0.502513	0.500250	0.500025

Approaching 1 from the right:

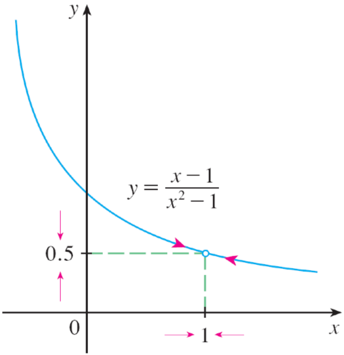
$x > 1$	1.5	1.1	1.01	1.001	1.0001
$f(x)$	0.400000	0.476190	0.497512	0.499750	0.499975

Example 1–Solution

On the basis of the values in the tables, we make the guess that

$$\lim_{x \to 1} \frac{x-1}{x^2-1} = \frac{1}{2}.$$

Example 1 is illustrated by the graph of f in the figure below.



Example 1–Solution

Now let's change f slightly by giving it the value 2 when $x = 1$ and calling the resulting function g:

$$g(x) = \begin{cases} \frac{x-1}{x^2-1}, & x \neq 1 \\ 2, & x = 1 \end{cases}$$

$g(x) \neq f(x) = \frac{x-1}{x^2-1}$

This new function g still has the same limit as x approaches 1. (See Figure below.)

$\lim_{x \to 1} \frac{x-1}{x^2-1}$

$= \lim_{x \to 1} \frac{x-1}{(x+1)(x-1)}$

$= \lim_{x \to 1} \frac{1}{x+1}$ $(\because x-1 \neq 0)$

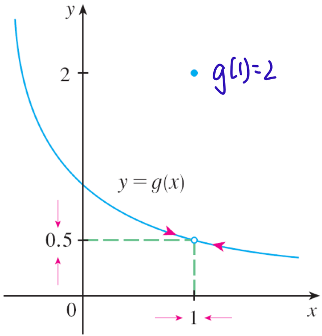
$\boxed{f(x) = \frac{x-1}{x^2-1} \quad h(x) = \frac{1}{x+1} \Rightarrow f(x) \neq h(x)}$

$\mathrm{Dom}(f) = \mathbb{R} \setminus \{-1, 1\}$

$\mathrm{Dom}(h) = \mathbb{R} \setminus \{-1\}$

$f(x)$ and $h(x)$ behave the same when $x \neq 1$

$\Rightarrow$ they should have same limit value as $x \to 1$.



Exercises:

Estimate the values of the following limits.

(a) $\lim_{t\to 0} \frac{\sqrt{t^2+16}-4}{t^2}$ (b) $\lim_{x\to 0} \frac{\sin 2x}{2x}$ (c) $\lim_{x\to 0} \sin\frac{\pi}{x}$

a) $\lim_{t\to 0} \frac{\sqrt{t^2+16}-4}{t^2}\left(\frac{\sqrt{t^2+16}+4}{\sqrt{t^2+16}+4}\right)$

$= \lim_{t\to 0} \frac{t^2}{t^2(\sqrt{t^2+16}+4)}$

$= \lim_{t\to 0} \frac{1}{\sqrt{t^2+16}+4}$ [$\because t\neq 0$

$= \frac{1}{8}$

b) $\lim_{x\to 0} \frac{\sin 2x}{2x} = 1$

Earlier:

$\lim_{x\to 0} \frac{\sin x}{x} = 1$

$\lim_{x\to 0} \frac{\sin 2x}{2x}$ let $t = 2x$

$\lim_{t\to 0} \frac{\sin t}{t} = 1$

$\therefore$ same!

c) $\lim_{x\to 0} \sin\left(\frac{\pi}{x}\right)$ does not exist

x	1	$\frac{1}{2}$	$\frac{1}{3}$	...	$\frac{1}{100}$
$\sin\frac{\pi}{x}$	0	0	0		0

For $x \to 0^+$

x	$-\frac{1}{2}$	$-\frac{1}{3}$	...	$-\frac{1}{100}$
$\sin\frac{\pi}{x}$	0	0		0

For $x \to 0^-$

We cannot say limit is 0!

This function don't approach a single value.

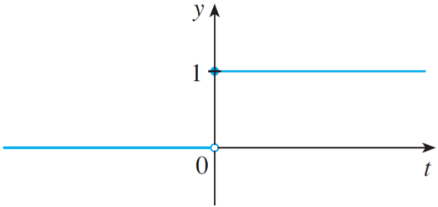
One-sided limits

Example 6

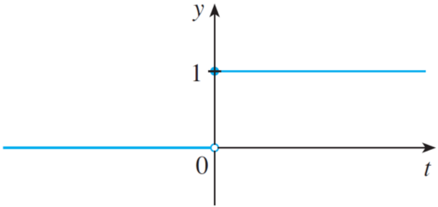
The Heaviside function H is defined by

$$H(\overset{t}{\cancel{x}}) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

This function is named after the electrical engineer Oliver Heaviside (1850-1925) and can be used to describe an electric current that is switched on at time $t = 0$. Its graph is shown below



Example 6



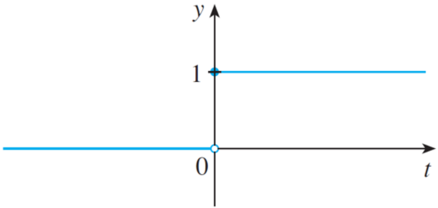
As t approaches 0 from the left, $H(t)$ approaches 0. As t approaches 0 from the right, $H(t)$ approaches 1. There is no single number that $H(t)$ approaches as t approaches 0. Therefore $\lim_{t \to 0} H(t)$ does not exist.

We noticed that $H(t)$ approaches 0 as t approaches 0 from the left and $H(t)$ approaches 1 as t approaches 0 from the right. We indicate this situation symbolically by writing

one sided limit

$$\lim_{t \to 0^-} H(t) = 0 \qquad \text{and} \qquad \lim_{t \to 0^+} H(t) = 1$$

Example 6



The notation $t \to 0^-$ indicates that we consider only values of t that are less than 0.

Likewise, the notation $t \to 0^+$ indicates that we consider only values of t that are greater than 0.

One-sided limits

Definition

We write

$$\lim_{x \to a^-} f(x) = L$$

and say the **left-hand limit of** $f(x)$ **as** x **approaches** a or the **limit of** $f(x)$ **as** x **approaches** a **from the left** is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be sufficiently close to a with x less than a.

$L-\varepsilon < f(x) < L+\varepsilon$ is required

try to find a working $(a-\delta, a)$ such that if $x \in (a-\delta, a)$, then $L-\varepsilon < f(x) < L+\varepsilon$

Similarly, if we require that x be greater than a, we get 'the **right-hand limit of** $f(x)$ **as** x **approaches** a is equal to L' and we write

$L-\varepsilon < f(x) < L+\varepsilon$

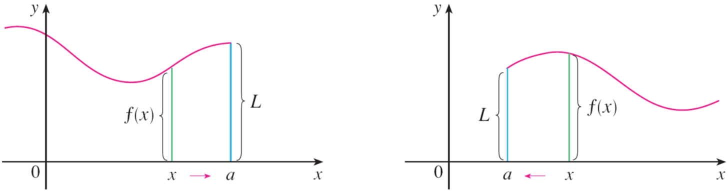
$$\lim_{x \to a^+} f(x) = L$$

$(a, a+\delta)$

The notation $x \to a^+$ means that we consider only x greater than a.

One-sided limits

The definitions of one-sided limits are illustrated in the following figure.



(a) $\lim_{x\to a^-} f(x) = L$ from left

(b) $\lim_{x\to a^+} f(x) = L$ from right

subset of

Theorem: The one-sided limits theorem

Let $f : D \to \mathbb{R}$ be a function. Then the limit $\lim_{x\to a} f(x)$ exists if and only if the two one-sided limits both exist and are equal. In that case

$$\lim_{x\to a} f(x) = \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x).$$

Exercise:

Consider the function

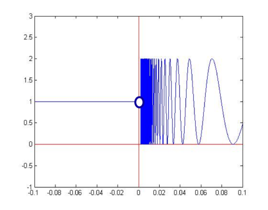
$$f(x) = \begin{cases} 1, & x < 0 \\ 1 + \sin\frac{1}{x}, & x > 0 \end{cases}$$

What do you expect the following limits to be?

(a) $\lim_{x\to 0^+} f(x)$ (b) $\lim_{x\to 0^-} f(x)$, $=1$ (c) $\lim_{x\to 0} f(x)$.

(a) does not approach a single value.
$\therefore$ does not exist

(c) does not exist
$\because$ the two one-sided limits are not equal



Intuitive definition of an infinite limit

Let f be a function defined on both sides of a, except possibly at a itself. Then

f(x) has no limit as $x \to 0$

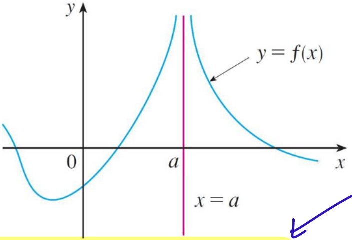
↳ $(a-\gamma, a) \cup (a, a+\gamma) \Rightarrow \gamma$ is a constant

$$\lim_{x \to a} f(x) = \infty$$

this means limit does not exist.

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a, but not equal to a.

restriction on a



$\infty + \infty = \infty$

$2\infty = \infty$

$2 = 1$

*does not mean the limit exists.

Comment: This does not mean that ∞ is a number. Nor does it mean that the limit exists. It simply expresses the particular way in which the limit does not exist: $f(x)$ can be made as large as we like by taking x close enough to a.

Infinite limits

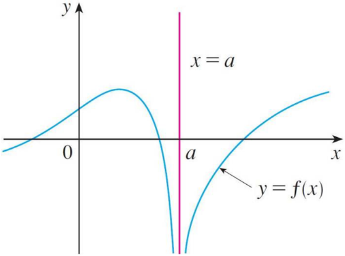
Definition

Let f be a function defined on both sides of a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = -\infty$$

f(x) does not have a limit as $x \to a$.

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a, but not equal to a.



Comment: $\lim_{x \to a} f(x) = -\infty$ can be read as *the limit of $f(x)$, as x approaches a, is negative infinity* or *$f(x)$ decreases without bound as x approaches a.*

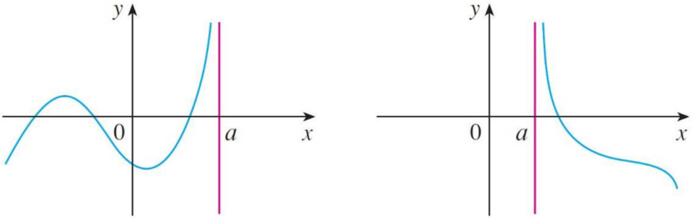
Similar definitions can be given for the one-sided infinite limits

$$\lim_{x\to a^-} f(x) = \infty \qquad \lim_{x\to a^+} f(x) = \infty$$

$$\lim_{x\to a^-} f(x) = -\infty \qquad \lim_{x\to a^+} f(x) = -\infty$$

The limits do not exist for these 4 functions.

remembering that $x \to a^-$ means that we consider only values of x that are less than a, and similarly $x \to a^+$ means that we consider only $x > a$.



(a) $\lim_{x\to a^-} f(x) = \infty$ (b) $\lim_{x\to a^+} f(x) = \infty$

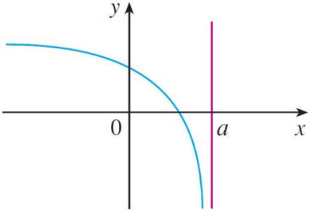
One-sided infinite limits (Cont'd)

Similar definitions can be given for the one-sided infinite limits

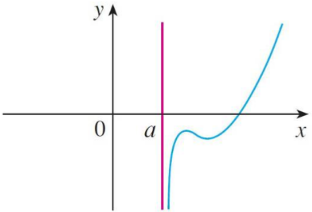
$$\lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

$$\lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$

remembering that $x \to a^-$ means that we consider only values of x that are less than a, and similarly $x \to a^+$ means that we consider only $x > a$.



(c) $\lim\limits_{x \to a^-} f(x) = -\infty$



(d) $\lim\limits_{x \to a^+} f(x) = -\infty$

Definition

The vertical line $x = a$ is called a vertical asymptote of the curve $y = f(x)$ if at least one of the following statements is true

$$\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

$$\lim_{x \to a} f(x) = -\infty \qquad \lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$

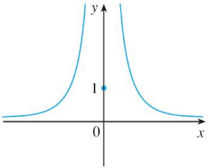
For instance, the line $x = 0$ is a vertical asymptote of the function

(0,1) is on the vertical asymptote.

$$y = f(x) = \begin{cases} \frac{1}{x^2}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

Dom(f) = $\mathbb{R}$

f is an even function.



Example 10

Find the vertical asmptotes of $f(x) = \tan x$.

$(2k+1)\pi + \frac{\pi}{2}$

if $\cos x \neq 0$, then $\tan x$ is finite

Because

In $[0, 2\pi]$, when $x = \frac{\pi}{2}$ or $\frac{3\pi}{2}$, $\cos x = 0$

$\therefore 2k\pi + \frac{\pi}{2}$ or $2k\pi + \frac{3\pi}{2}$ $(k = 0, \pm 1, \pm 2 \dots)$

$$\tan x = \frac{\sin x}{\cos x}$$

$(\because \cos x \neq 0)$

there are potential vertical asymptotes where $\underline{\cos x = 0}$.

$n\pi + \frac{\pi}{2}$ $(n \in \mathbb{Z})$

In fact, since $\cos x \to 0^+$ as $x \to (\pi/2)^-$ and $\cos x \to 0^-$ as $x \to (\pi/2)^+$, whereas $\sin x$ is positive (near 1) when x is near $\pi/2$, we have

$$\lim_{x \to (\pi/2)^-} \tan x = \infty \qquad \text{and} \qquad \lim_{x \to (\pi/2)^+} \tan x = -\infty$$

This shows that the line $x = \pi/2$ is a vertical asymptote.

Similar reasoning shows that the lines $x = \pi/2 + n\pi$, where n is an integer, are all vertical asymptotes of $f(x) = \tan x$.

$\sin(n\pi + \frac{\pi}{2}) = (-1)^n$

Example 10–Solution

The graph below confirms that the lines $x = \pi/2 + n\pi$, where n is an integer, are all vertical asymptotes of $f(x) = \tan x$.